University of St Andrews

ID5059 Coursework 2: Driver Scheduling for Ride-Hailing Service

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1. Introduction

The borough of Manhattan is a global trade hub and one of the key financial centers of the world with a growing technology scene, housing the majority of major US banks and tech company headquarters including IBM, Pepsico, and Verizon. In fact, Manhattan is estimated to have over 594 million square feet of office space [1]. It is also a tourist hub which attracted 65 million tourists in 2024 [2], serviced by its 3 major airports: LaGuardia, Newark, and JFK. Therefore, New York City is a bustling city serving both local commuters and international tourists.

The Taxi and Limousine Commission (TLC) published public datasets containing yellow and green taxi trip records for trips in New York City from 2009 [3]. This project uses the yellow taxi data from this dataset from 2024 only to predict taxi demand (by pickup and drop-off zones combined into a "Route") and average fare per trip using offline supervised regression models, including neural networks. We analyse what drives demand and contextualise these findings using average fare insights to provide a data-backed driver planning strategy. Our recommendation minimises passenger waiting times, avoids an excess of idle drivers, and maximises operating profit for the platform which is looking to launch operations in Manhattan. We assume that historical taxi trips (from 2024) are indicative of future taxi demand and fare-related information.

2. Data Cleaning and Preprocessing

The dataset initially contained approximately 41 million individual taxi trips across New York City's five boroughs, where each row represented a single journey. Each trip detailed geospatial information (pickup and drop-off locations), temporal data (date and time), and fare-related variables such as trip cost, tips, and surcharges.

To focus our analysis on Manhattan, all pickup locations were filtered to include Manhattan only, and drop-off locations across the five boroughs were included so long as the trip started in Manhattan. All outside-of-Manhattan drop-off locations were abstracted into a single "Outside" category in a derived *route* feature which describes in text the pickup ⟶ drop-off location of a taxi trip as a single feature. The 3 major airports (LaGuardia, Newark, JFK) were an exception and were kept as distinct routes, as we anticipated these locations to play an important role in our driver planning recommendation.

A taxi trip record was deemed invalid if the trip had a zero/negative fare or distance, or a drop-off time before its pickup time. These were assumed to be refunds or data errors, and would introduce noise into the predictive models, so were removed from the dataset. The rows (rides) were then aggregated, summed by pickup zone, drop-off zone, and time of day, so each row now describes all trips on a particular *route* (e.g., "Upper East Side South to Upper East Side North"), at a particular time (e.g., 16pm, which would include any trips between 16:00-16:59). This facilitated easier data exploration since the data could easily be aggregated into e.g. daily or hourly (*time_of_day*) trends, and the number of rows was reduced to ~4.3 million rows, a far more manageable data size. Additional data such as public holidays and weather information were included to provide extra "real-life context" to the models beyond the purely operational data provided.

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3. How well can we predict hourly demand per zone with the given data?

To predict hourly demand per zone we built 4 models:

- Model 1: Random Forest
- Model 2: Light Gradient-Boosting Machine (LightGBM)
- Model 3: Neural Network

To explain each model's results, the R2 score is used, a measure of the percentage of total variance being explained by the model. Put simply, it measures the difference between the predicted values and the actual values that were observed:

Model	Train	Validation	Test
Model 1: "Random Forest"	95.2%	88.0%	88.1%

Model 2: “LightGBM”	87.4%	87.3%	87.3%
Model 3: “Neural Network”	83.9%	84.1%	84.1%

Model selected for predictions: LightGBM

Model 1, Random Forest, is known to be particularly strong in making predictions when the data is very modular (ie is has many distinct features to use). This matches our dataset, where one may be trying to predict a trip at 12pm on a Monday from Upper East Side North to Upper East Side South, knowing that yesterday’s demand at 12pm was 400 on this route, there’s minimal cloud coverage and it is not a public holiday. Random forest uses each distinct feature to make decisions on its predictions, therefore the “depth” of the model (how many levels of decision it uses) was manually set to 25 since there are around 20 feature in the data. This is quite large, but means the model can find much smaller signals in the data. This is reflected in the model’s results, where Random Forest’s R2 score on the training set (subset of data used to train the model to make its predictions) was very high (95.2%). This means when using the Random Forest model, we can explain 95.2% of the variance of taxi trip data - a very accurate result. However, the drop in its performance on unseen data (88.0% and 88.1% for validation and test sets respectively) indicates that the model is potentially overfitting, meaning that it has learned the specific patterns in the training dataset rather than general patterns applicable to all taxi data (which is required to make reliable predictions).

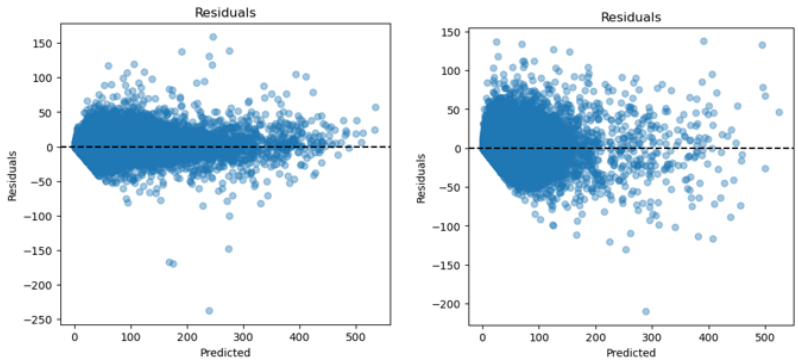

Figure 1: The performance of Model 1, Random Forest, on seen (left) vs unseen validation (right) data.

Model 2, LightGBM, took a fraction of the time to compute versus Models 1 (Random Forest) and 3 (Neural Network), making it particularly useful in analysis. It’s R2 score was also high, since it could explain 87.4% of the data’s variance in the training set, and 87.3% in both the validation and test set. While slightly slower than Model 1´s scores, the difference between the training, validation, and test scores is minimal, which indicates that this model is learning patterns at the right level of granularity unlike Model 1. It’s not too specific since it applies well to unseen data, but it’s also not abstracting over too much information since its test R2 score is still high at 87.3%. A combination of this model’s accurate and reliable predictions for taxi demand and its very efficient runtime led to this model being fine-tuned more extensively than others to make generate the final predictions for taxi demand.

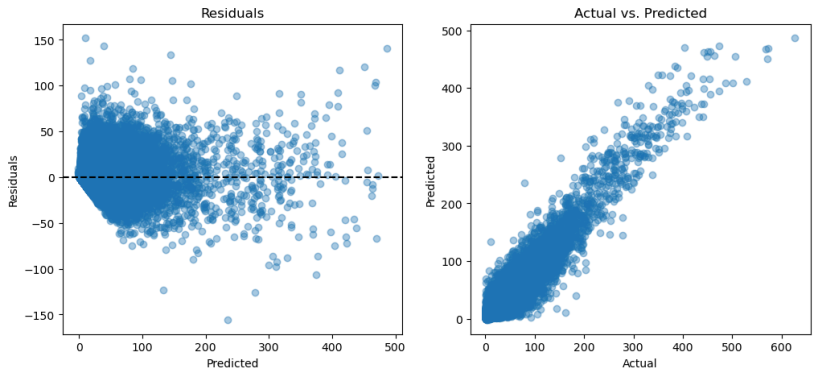

Figure 2: The performance of LightGBM on the test (unseen) data, where a straight line in the right-hand graph indicates strong predictive accuracy (R2 score).

Model 3, the Neural Network, was the most computationally-expensive model which implemented a deep learning approach. Despite this, it returned the worst results across all datasets (train, validation, and test). However, we note that this may be due to a limitation of our lack of computing power. Increasing the number of *max_iterations* for this model would allow the algorithm more time to compute, and every time this variable was increased during fine-tuning, all R2 scores across every dataset increased. Given the R2 scores didn't plateau even with a very large number of iterations (largest tried was 2000), this model is still very promising. If we were expanding this project into a production-ready prediction model, it may be that this model actually obtains the best results, if specialist compute/hardware can be provided.

3.1. What drives demand?

3.1.1. Pickup and Drop-Off Zones

Upper East Side (North and South), dominates demand in pickup zones, but also routes; the 4 biggest routes are Upper East Side South to South, South to North, North to South, and North to North. The center of Manhattan is the core of business throughout the year, with Midtown second to Upper East Side.

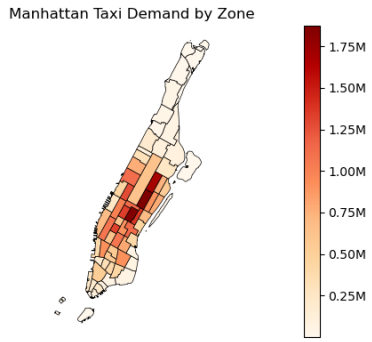

Figure 3: Ride demand by Manhattan zones, using pickup as location

3.1.2. Day of the Week

The patterns here perhaps describe a typical commuter's weekly calendar: Monday consistently has the lowest demand (remote work, for example), and Sunday has the second lowest. Meanwhile, Thursday and Saturday compete for the highest demand days. Accounting for public holidays, this trend is almost always true, throughout the year.

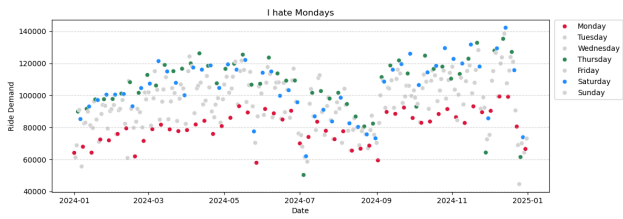

Figure 4: Daily ride demand, coloured by day of the week

3.1.3. Time of Day

Demand showed a clear trend depending on the time of day, and was bucketed into 5 groups to manage the data size while preserving the pattern across the day. It also made for a more sensible driver planning recommendation, as explained in Section 5.

"Late Night": 00:00 – 05:59
"Morning Rush": 06:00 – 09:59
"Midday": 10:00 – 15:59
"Evening Rush": 16:00 – 19:59
"Late Evening": 20:00 – 23:59

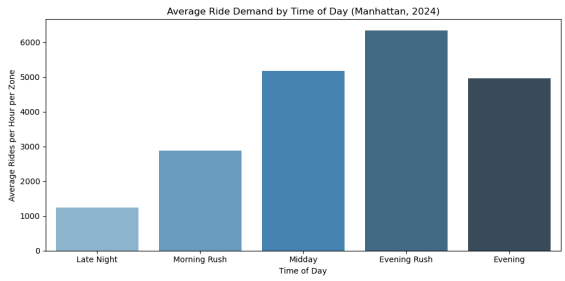

Figure 5: Average daily ride demand, bucketed by 5 times of day

These time periods are also visible in the heatmap of hourly demand over the week.

3.1.4. Demand Yesterday (at same time of day)

The data shows a consistent trend in daily demand across the days of the week. Meaning, the demand for taxi trips 24 hours ago is a very good predictor for what the demand for taxi trips will be like now. This can be shown by the clear vertical colour pattern in the red/white graph on the ride-hand side, where each square has a relatively similar colour to the square above it (24 hours before).

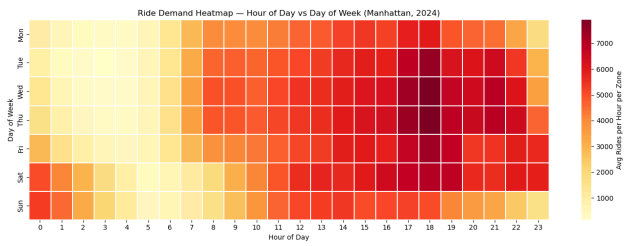

Figure 6: Heat map of average ride demand by hour (x-axis) and day of the week (y-axis)

3.1.5. Weather

Cloud coverage and precipitation level often increases demand, but do not cause anomalously high values on their own. Temperature describes the annual trend well - i.e., extreme values represent summer & Christmas where there is reduced demand. However, temperature is likely not driver of demand, but represents seasonal calendar being a driver.

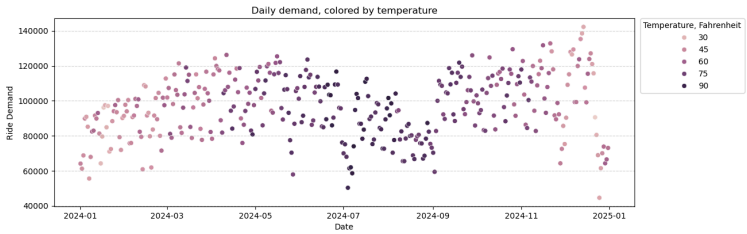

Figure 7: Total daily demand, coloured by max temperature

3.1.6. Public Holidays

The daily demand experiences some unusually low data points that lie below the general M-shaped trend. Public holidays can be seen to explain away almost every one of these points. It is worth noting, this assumes supply is not the issue, but demand is - i.e., there are enough taxi drivers not taking the holidays to absorb supply.

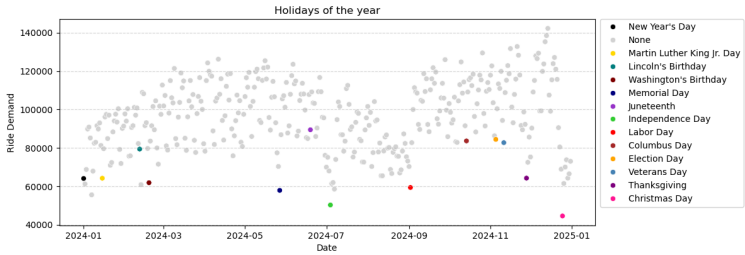

Figure 8: Total daily demand, coloured by public holiday

4. How well can we predict average fare per zone with the given data?

We prioritised average fare over total revenue to better align with the demand findings and provide an actionable strategy in Section 5. While hourly revenue provides a broad overview, it doesn't distinguish between purely high-volume demand and actual high-value areas (high fare AND high demand). Modeling average fare clarifies this distinction, allowing for more precise driver allocation and reduced idle time.

To predict average fare per zone we built 3 models:

- Model 1: Linear Regression
- Model 2: MLP Neural Network
- Model 3: LightGBM

Model	Train	Validation	Test
Model 1: "Linear Regression"	33.5%	35.7%	31.1%
Model 2: "LightGBM"	82.3%	84.0%	73.4%
Model 3: "Neural Network"	68.6%	70.3%	61.3%

Model selected for predictions: LightGBM

Model 1, Linear Regression, demonstrates the weakest performance among the models for average fare prediction. It achieves R2 values between 31.1% and 35.7%. The purpose of this model is to disprove any notion that average fare may have a linear relationship. If there was a strong linear correlation, then a classical statistical approach be better suited, however, with such poor results, we can be certain that this dataset is nonlinear. Therefore, we continued to explore more advanced model algorithms.

Model 2, the Neural Network, appears capable of capturing these nonlinear patterns. R2 scores ranged between 61.3% and 70.3% - a massive increase from the linear model. Again, lack of computation is a limitation, since a plateau in performance was never observed, suggesting that this model could achieve greater results if the variable `max_iteration` is increased. Noticeably, the test set returned a huge decrease in accuracy. Looking at the predictions against the true values (Figure 9 below), it can be seen the test set (randomly assigned values from the total dataset) contains a singular data point with a huge average fare - well over 4000, where the mean average fare is around 20. Since R2 is a measure of the total distance between truth and prediction, this singular data point is skewing the test performance. This appears to be true for all 3 models.

Model 3, LightGBM, increases accuracies even further, with R2 scores between 73.4% - 84.0%, and, again, provides the most efficient computation. Given this model applies the decision tree methodology, the success rate highlights the data's predictability. As with almost every model, the lag feature (using yesterday's fare to predict today's fare) is the most effective. Even though the test set experiences a large drop-off in accuracy (due to this extremely large data point appearing in the test set), the consistency between the training accuracy, 82.3%, and validation accuracy, 84.0%, is reassuring. This suggests the model is generalizing effectively, and therefore there is little chance of overfitting - i.e., the model should continue to predict at similar rates on unseen data.

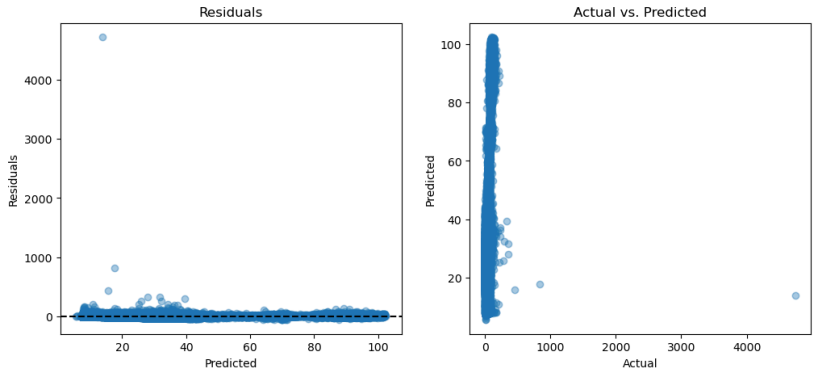

Figure 9: LightGBM test results. Error sizes versus predictions (left), and predictions versus true values (right). In both visuals, the unusually large average fare can be seen skewing the data

4.1. What drives average fare?

4.1.1. Time of Year (Month)

Average taxi fares are lower at the start of the year, presumably due to reduced tourism and a higher proportion of short, routine commuter trips. This rises as tourism peaks around May/June, and dips in July/August as commuters may begin to work more flexibly during peak summer or take their summer holiday. From September until the end of the year, fares are consistently higher on average, representing a combination of strong tourism and business activity, especially around Christmas in December.

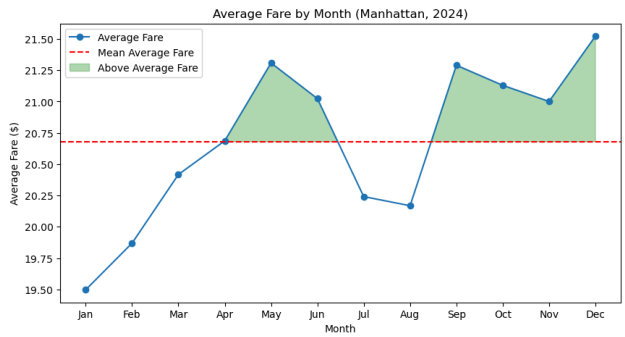

Figure 10: Average Fare across all Manhattan zones for each Month of the Year in 2024

4.1.2. Day of Week

Average fares steadily increase through the week, peaking on Thursday. This is likely driven by a mix of commuter travel and social activities such as after-work outings which typically happen on this day. Fares dip on Friday and fall below average over the weekend, potentially reflecting reduced commuter demand and greater use of alternative, more flexible transport options since people may have more free time.

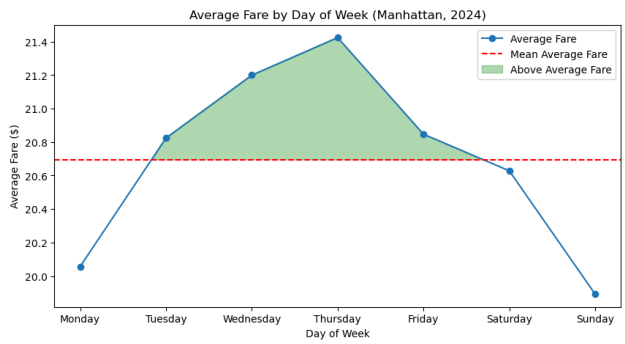

Figure 11: Average Fare across all Manhattan zones for each day of the week

4.1.3. Time of Day

While time of day was used in the models (as discussed above), hourly data was explored to highlight the time of day trends on a more granular level. There is a large average fare peak between 3am-6am before the working day starts. This may include people returning from evening plans or

commuting early for special reasons (eg a flight). While this is an off-peak time to travel, not many drivers would want to work at this time, explaining a peak in prices. Otherwise, aside from noticeable fare decreases during peak commuting hours (7-10am and 18-20pm), potentially due to lots of competition amongst drivers for passengers, the average fare is relatively stable throughout the day.

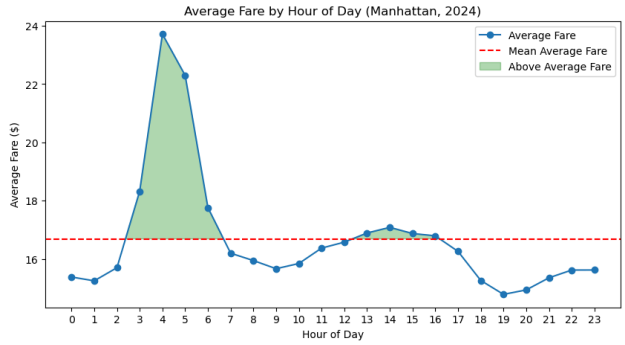

Figure 12: Average Fare across all Manhattan zones for each hour of the day

4.1.4. Public Holidays

Average fares were expected to be higher on public holidays due to assumed increased demand and lower driver supply. However, this was not the case as can be seen in the diagram where the average fare for public holidays (in coloured dots) are always on the lower half of the graph, indicating they are in fact lower. This could potentially be due to the workforce having the day off, and so choosing personal modes of transport to travel or not commuting at all to spend time with friends and family drives demand (and hence fare) down.

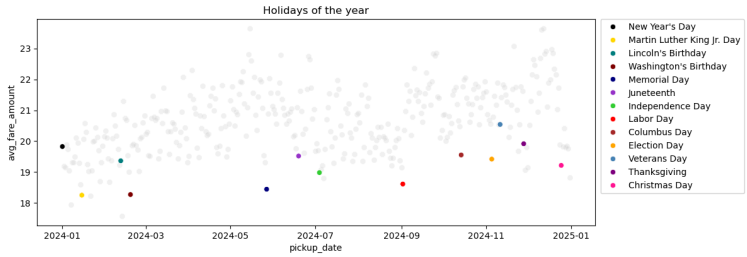

Figure 13: Average fares for public holidays in 2024

5. Profitability Insights and Driver Planning Recommendation

For our driver planning recommendation, our goals were to avoid idle drivers, minimise passenger waiting times, and maximise operating profit. Each of these can be tackled using our data-driven strategy:

1. **Avoid idle drivers:** identify which zones are high demand and when
2. **Minimise passenger waiting times:** ensure high demand zones are anticipated and covered by drivers
3. **Maximise operating profit:** ensure we get the right balance between high demand areas and high fare areas

We used time of day buckets instead of hourly data in our models to provide drivers with a reasonable amount of flexibility. In Manhattan, it is arguably unrealistic to direct a set amount of drivers to a particular place within an hour, since traffic and distance limitations may prevent this from being possible, causing operational difficulties. We use hourly demand/average fare in our graphs to clearly show the trends over the course of the day. An increasing number of drivers should be requested as demand grows from 7am and peaks at 18pm, and high priority should be placed around midday (10am-16pm) where both demand and average fare are high, making it a highly profitable timeframe. Even though the average fare is very high relative to other times, not many drivers should be deployed late at night (between midnight and 5am), since the demand is the lowest at this time.

We also consider driver location, and recommend sending a larger portion of drivers to Upper East Side and Midtown East, as these locations are geographically very close, are both in high demand, and both have relatively high averages fares as well, making making them highly profitable areas for the platform. During periods of strained driver availability, moving to Upper West Side (South) and East Chelsea is recommended, since the demand across many zones is similar, however the average fares in these particular areas are much higher.

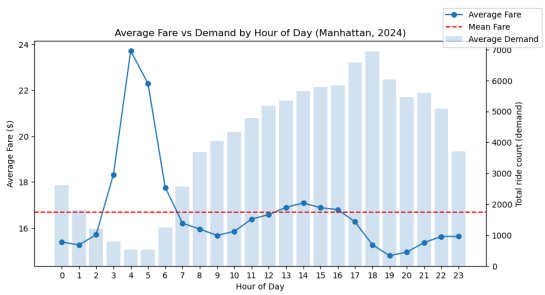

Figure 14: Demand and Fare Insights per Hour

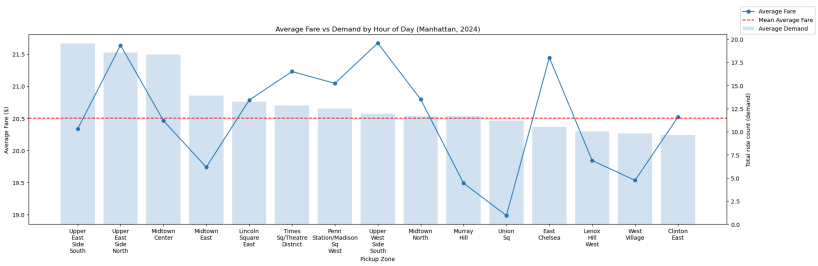

Figure 15: Demand and Fare Insights for the Top 15 Zones (in demand)

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[1] Office of the New York City Comptroller, “Spotlight: New York City's Office Market,” technical report, May 2024. [Online]. Available: https://comptroller.nyc.gov/reports/spotlight-new-york-citys-office-market/

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